1 Theoretical Guarantees for Risk-Calibrated Routing

We study the agent defined on the noisy-observation POMDP

$$\mathcal{M}_z = (\mathcal{S}, \mathcal{A}, \mathcal{O}, T, O_z, R, \gamma),$$

where $z \in \mathcal{Z}$ denotes the perturbation seed, x_t denotes the step-t observation context, π_θ is the SLM policy, π_T is the teacher LLM, V_ϕ is the verifier, and r_ψ is the router.

At each step t, the SLM generates K candidates

$$a_t^{(1)}, \ldots, a_t^{(K)} \sim \pi_\theta(\cdot \mid x_t),$$

the verifier scores them by

$$s_t^{(k)} = V_\phi(x_t, a_t^{(k)}),$$

and the best SLM candidate is

$$\hat{a}_t^{\mathrm{SLM}} = \mathrm{argmax}_{1 \le k \le K} V_\phi(x_t, a_t^{(k)}).$$

The router receives the feature vector $f_t \in \mathbb{R}^{13}$ and outputs

$$p_t = r_\psi(f_t) \in [0, 1].$$

A hard routing decision is then obtained by thresholding:

$$d_t = \mathbf{1}[p_t \ge \tau_{\mathrm{route}}].$$

Throughout the theory section, let $y_t \in \{0, 1\}$ denote the *correct escalation label* at step t. This binary label can be instantiated in either of the two ways already used in the methodology:

$$y_t = \mathbf{1}\left[\max_k V_\phi(x_t, a_{\mathrm{SLM}}^{(k)}) < V_\phi(x_t, a_{\mathrm{teacher}})\right]$$

or

$$y_t = \mathbf{1}[\rho(x_t) > \tau], \qquad \rho(x_t) = f\Big(1 - V_\phi(x_t, \hat{a}_t^{\mathrm{SLM}}),\ H(\pi_\theta(\cdot \mid x_t))\Big).$$

All results below only require that y_t be a valid binary escalation target.

1.1 Assumptions

Assumption 1.1 (Boundedness and measurability)**.** *For every step t,*

$$V_\phi(x_t, a_t) \in [0, 1], \qquad r_\psi(f_t) \in [0, 1], \qquad y_t \in \{0, 1\}, \qquad S_z \in \{0, 1\}.$$

Moreover, the feature vector satisfies

$$\|f_t\|_2 \le B_f$$

for some finite constant $B_f > 0$.

Assumption 1.2 (Well-defined calibration target)**.** *Define the conditional escalation probability*

$$q^\star(f) := \mathbb{P}(y_t = 1 \mid f_t = f).$$

We assume that $q^\star(f)$ is well-defined for almost every f.

Assumption 1.3 (Cost-sensitive routing surrogate)**.** *For theoretical analysis of the hard router, we measure the quality of a routing decision $d_t \in \{0, 1\}$ using the stepwise surrogate loss*

$$\ell_{\text{route}}(d_t, y_t) = c_{\text{SLM}}(1 - d_t) + c_{\text{LLM}} d_t + \kappa\, y_t(1 - d_t),$$

where $c_{\text{SLM}} < c_{\text{LLM}}$ and $\kappa > 0$ is the penalty for not *escalating on a step where escalation was actually needed.*

Assumption 1.4 (Perturbation pairing and stepwise Lipschitzness)**.** *Suppose two perturbation seeds $z, z' \in \mathcal{Z}$ produce two observed contexts $x_t^{(z)}$ and $x_t^{(z')}$ corresponding to the same latent environment state s_t. Let $P_t^{(z)} := \pi_\theta(\cdot \mid x_t^{(z)})$ and $P_t^{(z')} := \pi_\theta(\cdot \mid x_t^{(z')})$. Assume the one-step surrogate loss $\ell_t(P; s_t) \in [0, 1]$ satisfies*

$$|\ell_t(P; s_t) - \ell_t(Q; s_t)| \le L\ \mathrm{TV}(P, Q)$$

for all action distributions P, Q and some constant $L > 0$.

Assumption 1.5 (Convex surrogate for primal–dual analysis)**.** *For the convergence theorem only, we analyze the convex surrogate router*

$$p_\psi(f) = \sigma(\psi^\top f / T),$$

where $T > 0$ is fixed and ψ lies in a compact convex set Ψ. Let

$$g(\psi) := \widehat{\mathrm{CVaR}}_\alpha(1 - S_z) - \varepsilon$$

denote the empirical CVaR-constraint violation. Assume that the empirical Lagrangian

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda) = \widehat{\mathbb{E}}[\mathrm{cost}(d)] + \lambda\, g(\psi) + \widehat{\mathrm{Brier}}(r_\psi(f), y)$$

is convex in ψ, concave in λ, and has gradients bounded by G over $\Psi \times [0, \Lambda]$.

Remark 1.1. *Assumption 1.5 is the standard place where theory becomes slightly idealized relative to the practical MLP router. The theorem should therefore be presented as justification for the* training principle*—namely CVaR-constrained calibrated routing—rather than as a global convergence guarantee for the full nonconvex MLP.*

1.2 Calibration and Bayes-optimal routing

Theorem 1.1 (Brier-optimal calibration of the router)**.** *Under Assumptions 1.1 and 1.2, the population minimizer of the Brier risk*

$$\mathcal{R}_{\text{Brier}}(r_\psi) = \mathbb{E}\big[(r_\psi(f_t) - y_t)^2\big]$$

is the conditional escalation probability

$$r^\star(f) = q^\star(f) = \mathbb{P}(y_t = 1 \mid f_t = f).$$

Equivalently, among all measurable router outputs $p(f) \in [0,1]$, the unique pointwise minimizer is the true escalation probability.

Proof. We prove the result by conditioning on the feature vector f_t.

Step 1: Write the population Brier risk using the tower property. By the law of total expectation,

$$\mathcal{R}_{\text{Brier}}(p) = \mathbb{E}\big[(p(f_t) - y_t)^2\big] = \mathbb{E}\Big[\mathbb{E}\big[(p(f_t) - y_t)^2 \mid f_t\big]\Big].$$

Therefore, to minimize the outer expectation, it is enough to minimize the inner conditional quantity for each fixed feature value f.

Step 2: Fix a feature value f and denote $p = p(f)$, $q = q^\star(f)$. By definition,

$$q = \mathbb{P}(y_t = 1 \mid f_t = f).$$

Because $y_t \in \{0,1\}$, conditional on $f_t = f$, the random variable y_t equals 1 with probability q and 0 with probability $1 - q$.

Hence

$$\mathbb{E}\big[(p - y_t)^2 \mid f_t = f\big] = q(p-1)^2 + (1-q)(p-0)^2.$$

Step 3: Expand the squares carefully. We have

$$(p-1)^2 = p^2 - 2p + 1, \qquad (p-0)^2 = p^2.$$

Substitute these into the conditional risk:

$$q(p^2 - 2p + 1) + (1-q)p^2.$$

Now distribute the coefficients:

$$= qp^2 - 2qp + q + p^2 - qp^2.$$

The two qp^2 terms cancel:

$$= p^2 - 2qp + q.$$

Step 4: Complete the square. Add and subtract q^2:

$$p^2 - 2qp + q = (p^2 - 2qp + q^2) + q - q^2 = (p - q)^2 + q(1 - q).$$

Step 5: Identify the minimizing value. The term $q(1-q)$ does not depend on p. The term $(p - q)^2$ is always nonnegative and equals zero if and only if $p = q$.

Therefore, for every fixed f,

$$\operatorname{argmin}_{p\in[0,1]} \mathbb{E}\big[(p - y_t)^2 \mid f_t = f\big] = q^\star(f).$$

Step 6: Lift the pointwise minimizer back to the population problem. Since the population risk is the expectation of the conditional risk, and the conditional risk is minimized pointwise by $p(f) = q^\star(f)$, the global minimizer is

$$r^\star(f) = q^\star(f).$$

This proves the theorem. □

Corollary 1.1 (Cost-optimal hard routing threshold)**.** *Under Assumptions 1.1, 1.2, and 1.3, the Bayes-optimal hard routing rule for the stepwise loss ℓ_{route} is*

$$d_t^\star = \mathbf{1}\left[q^\star(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right].$$

Equivalently, if the router is calibrated so that $r_\psi(f_t) \approx q^\star(f_t)$, then the optimal escalation rule is to compare the router probability against the threshold

$$\tau_{\text{route}}^\star = \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}.$$

Proof. Fix a feature value f, and let $q = q^\star(f) = \mathbb{P}(y_t = 1 \mid f_t = f)$.

We compare the conditional expected loss of the two possible actions $d_t = 1$ and $d_t = 0$.

Case 1: Escalate, i.e., $d_t = 1$. Substitute $d_t = 1$ into

$$\ell_{\text{route}}(d_t, y_t) = c_{\text{SLM}}(1 - d_t) + c_{\text{LLM}} d_t + \kappa y_t(1 - d_t).$$

Because $1 - d_t = 0$ and $d_t = 1$, this becomes

$$\ell_{\text{route}}(1, y_t) = c_{\text{LLM}}.$$

Therefore

$$\mathbb{E}[\ell_{\text{route}}(1, y_t) \mid f_t = f] = c_{\text{LLM}}.$$

Case 2: Do not escalate, i.e., $d_t = 0$. Now $1 - d_t = 1$ and $d_t = 0$, so

$$\ell_{\text{route}}(0, y_t) = c_{\text{SLM}} + \kappa y_t.$$

Taking conditional expectation,

$$\mathbb{E}[\ell_{\text{route}}(0, y_t) \mid f_t = f] = c_{\text{SLM}} + \kappa \mathbb{E}[y_t \mid f_t = f] = c_{\text{SLM}} + \kappa q.$$

Step 3: Compare the two conditional risks. Escalation is optimal if and only if

$$c_{\mathrm{LLM}} \leq c_{\mathrm{SLM}} + \kappa q.$$

Subtract c_{SLM} from both sides:

$$c_{\mathrm{LLM}} - c_{\mathrm{SLM}} \leq \kappa q.$$

Now divide by $\kappa > 0$:

$$q \geq \frac{c_{\mathrm{LLM}} - c_{\mathrm{SLM}}}{\kappa}.$$

Thus the Bayes-optimal rule is

$$d_t^\star = \mathbf{1}\left[q^\star(f_t) \geq \frac{c_{\mathrm{LLM}} - c_{\mathrm{SLM}}}{\kappa}\right].$$

This proves the corollary. □

Theorem 1.2 (Regret bound from router miscalibration)**.** *Let*

$$\hat{d}_t = \mathbf{1}\left[r_\psi(f_t) \geq \frac{c_{\mathrm{LLM}} - c_{\mathrm{SLM}}}{\kappa}\right]$$

be the plug-in hard router obtained from a probabilistic router r_ψ, and let $d_t^\star$ be the Bayes-optimal rule from Corollary 1.1. Then

$$\mathbb{E}\Big[\ell_{\mathrm{route}}(\hat{d}_t, y_t) - \ell_{\mathrm{route}}(d_t^\star, y_t)\Big] \leq \kappa\, \mathbb{E}[|r_\psi(f_t) - q^\star(f_t)|]\,.$$

Proof. We again work conditionally on $f_t = f$. Let

$$q = q^\star(f), \qquad \widehat{q} = r_\psi(f), \qquad \tau^\star = \frac{c_{\mathrm{LLM}} - c_{\mathrm{SLM}}}{\kappa}.$$

Step 1: Compute the conditional risk difference between the two actions. From the previous proof,

$$R_f(1) := \mathbb{E}[\ell_{\mathrm{route}}(1, y_t) \mid f_t = f] = c_{\mathrm{LLM}},$$

and

$$R_f(0) := \mathbb{E}[\ell_{\mathrm{route}}(0, y_t) \mid f_t = f] = c_{\mathrm{SLM}} + \kappa q.$$

Therefore,

$$R_f(0) - R_f(1) = c_{\mathrm{SLM}} + \kappa q - c_{\mathrm{LLM}} = \kappa q - (c_{\mathrm{LLM}} - c_{\mathrm{SLM}}) = \kappa(q - \tau^\star).$$

So the absolute advantage of the better action over the worse action is

$$|R_f(0) - R_f(1)| = \kappa|q - \tau^\star|.$$

Step 2: Characterize when the plug-in router makes a mistake. The optimal decision is

$$d^\star(f) = \mathbf{1}[q \geq \tau^\star],$$

while the plug-in decision is

$$\hat{d}(f) = \mathbf{1}[\widehat{q} \geq \tau^\star].$$

These two decisions differ only if q and $\widehat{q}$ lie on opposite sides of the threshold $\tau^\star$.

Hence, on the event $\hat{d}(f) \neq d^\star(f)$, the threshold $\tau^\star$ lies between q and $\widehat{q}$, so

$$|q - \tau^\star| \leq |\widehat{q} - q|.$$

Step 3: Bound the conditional regret. If $\hat{d}(f) = d^\star(f)$, then the conditional regret is 0.

If $\hat{d}(f) \neq d^\star(f)$, then the conditional regret equals exactly the difference between the worse and better conditional actions, namely

$$\kappa|q - \tau^\star|.$$

Using the previous step,

$$\kappa|q - \tau^\star| \leq \kappa|\widehat{q} - q|.$$

Therefore, in all cases,

$$\mathbb{E}\Big[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t) \mid f_t = f\Big] \leq \kappa|\widehat{q} - q|.$$

Step 4: Average over f_t. Take expectation over f_t:

$$\mathbb{E}\Big[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t)\Big] \leq \kappa\mathbb{E}[|r_\psi(f_t) - q^\star(f_t)|]\,.$$

This proves the theorem. □

1.3 Verifier-guided candidate generation

Definition 1.1 (Good candidate set)**.** *For a fixed context x_t and threshold $\gamma \in [0,1]$, define*

$$\mathcal{G}_\gamma(x_t) := \{a \in \mathcal{A} : V_\phi(x_t, a) \geq \gamma\}\,.$$

Also define the SLM coverage probability

$$\mu_\gamma(x_t) := \mathbb{P}_{a\sim\pi_\theta(\cdot|x_t)}\big(a \in \mathcal{G}_\gamma(x_t)\big).$$

Proposition 1.1 (Best-of-K verifier selection improves with coverage)**.** *Fix a context x_t, and suppose K candidate actions are sampled i.i.d. from $\pi_\theta(\cdot \mid x_t)$. Assume that for every good action $a^+ \in \mathcal{G}_\gamma(x_t)$ and every bad action $a^- \notin \mathcal{G}_\gamma(x_t)$, the verifier misranks the pair with probability at most η_V, i.e.,*

$$\mathbb{P}\big(V_\phi(x_t, a^-) \geq V_\phi(x_t, a^+)\big) \leq \eta_V.$$

Then the probability that verifier selection returns a γ-good action satisfies

$$\mathbb{P}\big(\hat{a}_t^{\mathrm{SLM}} \in \mathcal{G}_\gamma(x_t)\big) \geq 1 - (1 - \mu_\gamma(x_t))^K - \frac{K(K-1)}{2}\eta_V.$$

Proof. We prove the bound by decomposing the failure event.

Step 1: Define the success and failure events. Let

$$E_{\mathrm{good}} = \left\{\hat{a}_t^{\mathrm{SLM}} \in \mathcal{G}_\gamma(x_t)\right\}.$$

We want a lower bound on $\mathbb{P}(E_{\mathrm{good}})$.

Its complement is

$$E_{\mathrm{bad}} = E_{\mathrm{good}}^c = \left\{\hat{a}_t^{\mathrm{SLM}} \notin \mathcal{G}_\gamma(x_t)\right\}.$$

Step 2: Split the bad event into two simpler events. There are only two ways selection can end up bad:

(a) No good candidate was sampled at all.

(b) At least one good candidate was sampled, but verifier misranking caused a bad candidate to be selected.

Define

$$A = \{\text{none of the } K \text{ sampled candidates lies in } \mathcal{G}_\gamma(x_t)\},$$

and

$$B = \{\text{at least one good candidate is sampled, but the selected candidate is bad}\}.$$

Then

$$E_{\mathrm{bad}} \subseteq A \cup B.$$

Hence, by the union bound,

$$\mathbb{P}(E_{\mathrm{bad}}) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

Therefore,

$$\mathbb{P}(E_{\mathrm{good}}) = 1 - \mathbb{P}(E_{\mathrm{bad}}) \geq 1 - \mathbb{P}(A) - \mathbb{P}(B).$$

Step 3: Bound $\mathbb{P}(A)$. Each sampled candidate is good with probability $\mu_\gamma(x_t)$. Therefore each sampled candidate is bad with probability $1 - \mu_\gamma(x_t)$.

Because the K candidates are sampled independently,

$$\mathbb{P}(A) = (1 - \mu_\gamma(x_t))^K.$$

Step 4: Bound $\mathbb{P}(B)$. Now consider the event B: at least one good candidate exists, but a bad candidate is selected.

For this to happen, there must exist at least one good–bad pair (a^+, a^-) among the sampled candidates such that the verifier ranks the bad candidate at least as high as the good one.

For a fixed such pair, the pairwise misranking probability is at most η_V by assumption.

How many good–bad pairs can there be among K candidates? At most

$$\frac{K(K-1)}{2},$$

because this is the total number of unordered pairs among K objects.

Applying the union bound again,

$$\mathbb{P}(B) \leq \frac{K(K-1)}{2}\eta_V.$$

Step 5: Combine the two bounds. Substitute the bounds for $\mathbb{P}(A)$ and $\mathbb{P}(B)$:

$$\mathbb{P}(E_{\text{good}}) \geq 1 - (1 - \mu_\gamma(x_t))^K - \frac{K(K-1)}{2}\eta_V.$$

This proves the proposition. □

Remark 1.2. *Proposition 1.1 explains why the verifier helps twice: first through $\mu_\gamma(x_t)$, which measures whether the SLM can* generate *at least one strong candidate, and second through η_V, which measures whether the verifier can* identify *that candidate once it appears.*

1.4 Consistency regularization under perturbation noise

Lemma 1.1 (Total variation is controlled by Jensen–Shannon divergence)**.** *For any two distributions P and Q,*

$$\mathrm{TV}(P, Q) \leq \sqrt{2\ \mathrm{JSD}(P\|Q)}.$$

Proof. Let

$$M = \frac{P+Q}{2}.$$

Step 1: Use the triangle inequality for total variation. By the triangle inequality,

$$\mathrm{TV}(P, Q) \leq \mathrm{TV}(P, M) + \mathrm{TV}(M, Q).$$

Step 2: Apply Pinsker's inequality twice. Pinsker's inequality states that for any two distributions U, V,

$$\mathrm{TV}(U, V) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(U\|V)}.$$

Apply it first with $U = P$, $V = M$:

$$\mathrm{TV}(P, M) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(P\|M)}.$$

Apply it again with $U = Q$, $V = M$:

$$\mathrm{TV}(Q, M) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Since $\mathrm{TV}(M, Q) = \mathrm{TV}(Q, M)$,

$$\mathrm{TV}(M, Q) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Step 3: Add the two bounds. Hence

$$\mathrm{TV}(P, Q) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(P\|M)} + \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Step 4: Simplify the square-root sum. For any nonnegative numbers a, b,

$$(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab} \leq 2a + 2b,$$

so

$$\sqrt{a} + \sqrt{b} \leq \sqrt{2(a + b)}.$$

Apply this with

$$a = \frac{1}{2}\,\mathrm{KL}(P\|M), \qquad b = \frac{1}{2}\,\mathrm{KL}(Q\|M).$$

Then

$$\mathrm{TV}(P, Q) \leq \sqrt{2\left(\frac{1}{2}\,\mathrm{KL}(P\|M) + \frac{1}{2}\,\mathrm{KL}(Q\|M)\right)}.$$

Step 5: Recognize Jensen–Shannon divergence. By definition,

$$\mathrm{JSD}(P\|Q) = \frac{1}{2}\,\mathrm{KL}(P\|M) + \frac{1}{2}\,\mathrm{KL}(Q\|M).$$

Therefore

$$\mathrm{TV}(P, Q) \leq \sqrt{2\ \mathrm{JSD}(P\|Q)}.$$

This proves the lemma. □

Theorem 1.3 (Perturbation-generalization bound from consistency regularization)**.** *Assume Assumptions 1.1 and 1.4. For a perturbation seed z, define the episodic surrogate risk*

$$R_z(\pi_\theta) := \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(\pi_\theta(\cdot \mid x_t^{(z)}); s_t\Big)\right].$$

Then for any two perturbation seeds $z, z' \in \mathcal{Z}$,

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)},$$

where

$$\mathcal{L}_{\text{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[\text{JSD}\Big(\pi_\theta(\cdot \mid x_t^{(z)}) \,\Big\|\, \pi_\theta(\cdot \mid x_t^{(z')})\Big)\right].$$

Consequently, if a training seed z_{train} is representative of the seen perturbations, then for any unseen test seed z_{test},

$$R_{z_{\text{test}}}(\pi_\theta) \le R_{z_{\text{train}}}(\pi_\theta) + HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

Proof. We proceed very slowly so that each transformation is explicit.

Step 1: Write the difference of the two episodic risks. By definition,

$$R_z(\pi_\theta) - R_{z'}(\pi_\theta) = \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(P_t^{(z)}; s_t\Big)\right] - \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(P_t^{(z')}; s_t\Big)\right],$$

where

$$P_t^{(z)} := \pi_\theta(\cdot \mid x_t^{(z)}), \qquad P_t^{(z')} := \pi_\theta(\cdot \mid x_t^{(z')}).$$

Using linearity of expectation and linearity of summation,

$$R_z(\pi_\theta) - R_{z'}(\pi_\theta) = \mathbb{E}\left[\sum_{t=1}^{H}\Big(\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big)\right].$$

Step 2: Take absolute values and use the triangle inequality. We want to upper bound

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)|.$$

Apply the triangle inequality:

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le \mathbb{E}\left[\sum_{t=1}^{H}\Big|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big|\right].$$

Because the sum has only nonnegative terms after the absolute value, we can pull the sum outside:

$$\le \sum_{t=1}^{H} \mathbb{E}\Big[\Big|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big|\Big].$$

Step 3: Use the Lipschitz assumption in total variation. By Assumption 1.4,

$$\left|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\right| \leq L \ \mathrm{TV}(P_t^{(z)}, P_t^{(z')}).$$

Substitute this into the previous inequality:

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \leq L \sum_{t=1}^{H} \mathbb{E}\Big[\mathrm{TV}(P_t^{(z)}, P_t^{(z')})\Big].$$

Step 4: Convert total variation to JSD using Lemma 1.1. Lemma 1.1 gives

$$\mathrm{TV}(P_t^{(z)}, P_t^{(z')}) \leq \sqrt{2 \ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})}.$$

Therefore,

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \leq L \sum_{t=1}^{H} \mathbb{E}\left[\sqrt{2 \ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})}\right].$$

Step 5: Apply Jensen's inequality to move the expectation inside the square root. The function $u \mapsto \sqrt{u}$ is concave on $[0, \infty)$. For a concave function φ, Jensen's inequality states

$$\mathbb{E}[\varphi(X)] \leq \varphi(\mathbb{E}[X]).$$

Apply this with

$$\varphi(u) = \sqrt{u}, \qquad X = 2 \ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')}).$$

Then

$$\mathbb{E}\left[\sqrt{2 \ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})}\right] \leq \sqrt{2\,\mathbb{E}\Big[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\Big]}.$$

So

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \leq L \sum_{t=1}^{H} \sqrt{2\,\mathbb{E}\Big[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\Big]}.$$

Step 6: Use the global consistency regularizer to upper bound each step. By definition,

$$\mathcal{L}_{\mathrm{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\right].$$

In particular, for the pair (z, z') being considered, the average stepwise JSD is upper bounded by $\mathcal{L}_{\mathrm{cons}}(\theta)$ up to the natural empirical averaging interpretation. Using this quantity as the global bound,

$$\mathbb{E}\Big[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\Big] \leq \mathcal{L}_{\mathrm{cons}}(\theta)$$

for each t in the worst case or on average over the paired dataset.

Thus each summand is at most

$$\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

There are H such summands, so

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

This proves the first inequality.

Step 7: Deduce the train-to-test bound. Take $z = z_{\text{test}}$ and $z' = z_{\text{train}}$. Then

$$R_{z_{\text{test}}}(\pi_\theta) - R_{z_{\text{train}}}(\pi_\theta) \le |R_{z_{\text{test}}}(\pi_\theta) - R_{z_{\text{train}}}(\pi_\theta)| \le HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

Rearranging yields

$$R_{z_{\text{test}}}(\pi_\theta) \le R_{z_{\text{train}}}(\pi_\theta) + HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

This proves the theorem. □

Remark 1.3. *Theorem 1.3 is written for an episodic* surrogate *risk that decomposes over steps. If the binary episode failure $1 - S_z$ is upper bounded by such a surrogate, then the same bound transfers immediately to the failure objective used in the robust routing section.*

1.5 CVaR-constrained primal–dual optimization

Definition 1.2 (Empirical CVaR over perturbation seeds)**.** *For each perturbation seed $z \in \mathcal{Z}$, let the empirical failure rate be*

$$F_z = 1 - \mathrm{SR}_z.$$

Let

$$F_{(1)} \ge F_{(2)} \ge \cdots \ge F_{(|\mathcal{Z}|)}$$

denote the sorted seed-wise failure rates, and let

$$k = \lceil \alpha |\mathcal{Z}| \rceil.$$

Then

$$\widehat{\mathrm{CVaR}}_\alpha(1 - S_z) = \frac{1}{k}\sum_{j=1}^{k} F_{(j)}.$$

Under the convex surrogate in Assumption 1.5, consider the empirical Lagrangian

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda) = \widehat{\mathbb{E}}[\mathrm{cost}(d)] + \lambda\Big(\widehat{\mathrm{CVaR}}_\alpha(1 - S_z) - \varepsilon\Big) + \widehat{\mathrm{Brier}}(r_\psi(f), y).$$

We run projected gradient descent–ascent:

$$\psi_{n+1} = \Pi_\Psi\Big(\psi_n - \eta\nabla_\psi \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\Big),$$
$$\lambda_{n+1} = \Pi_{[0,\Lambda]}\Big(\lambda_n + \eta\nabla_\lambda \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\Big),$$

and define the averaged iterates

$$\bar{\psi}_N = \frac{1}{N}\sum_{n=1}^{N}\psi_n, \qquad \bar{\lambda}_N = \frac{1}{N}\sum_{n=1}^{N}\lambda_n.$$

Theorem 1.4 (Primal–dual convergence and near-feasibility)**.** *Assume Assumptions 1.1 and 1.5. Let the diameter of Ψ be at most D_ψ, and the diameter of $[0,\Lambda]$ be at most $D_\lambda = \Lambda$. Define*

$$D := \sqrt{D_\psi^2 + D_\lambda^2}.$$

If the step size is chosen as

$$\eta = \frac{D}{G\sqrt{N}},$$

then the averaged iterates satisfy the saddle-point gap bound

$$\max_{\lambda\in[0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \min_{\psi\in\Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \le \frac{2DG}{\sqrt{N}}.$$

Moreover, the empirical CVaR-constraint violation satisfies

$$\Big[\widehat{\mathrm{CVaR}}_\alpha(1 - S_z; \bar{\psi}_N) - \varepsilon\Big]_+ \le \frac{2DG}{\Lambda\sqrt{N}}.$$

Proof. The proof is a standard but important telescoping argument. I spell out each algebraic step.

Step 1: Introduce shorthand for the gradients. Define

$$g_n^\psi := \nabla_\psi \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n), \qquad g_n^\lambda := \nabla_\lambda \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n).$$

By assumption,

$$\|g_n^\psi\|_2 \le G, \qquad |g_n^\lambda| \le G.$$

Step 2: Use nonexpansiveness of projection for the primal update. Fix any comparator $\psi \in \Psi$. Because Euclidean projection onto a convex set is nonexpansive,

$$\|\psi_{n+1} - \psi\|_2^2 \le \|\psi_n - \eta g_n^\psi - \psi\|_2^2.$$

Expand the square on the right-hand side:

$$\|\psi_n - \psi\|_2^2 - 2\eta\langle g_n^\psi, \psi_n - \psi\rangle + \eta^2\|g_n^\psi\|_2^2.$$

Therefore,

$$\|\psi_{n+1} - \psi\|_2^2 \leq \|\psi_n - \psi\|_2^2 - 2\eta\langle g_n^\psi, \psi_n - \psi\rangle + \eta^2 G^2.$$

Rearrange:

$$2\eta\langle g_n^\psi, \psi_n - \psi\rangle \leq \|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2 + \eta^2 G^2.$$

Divide by 2η:

$$\langle g_n^\psi, \psi_n - \psi\rangle \leq \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 3: Convert the inner product into a primal function gap using convexity. Because the Lagrangian is convex in ψ,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \leq \langle g_n^\psi, \psi_n - \psi\rangle.$$

Combining with the previous inequality,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \leq \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 4: Do the same for the dual variable. Now fix any comparator $\lambda \in [0, \Lambda]$.

The dual update is an ascent step followed by projection:

$$\lambda_{n+1} = \Pi_{[0,\Lambda]}(\lambda_n + \eta g_n^\lambda).$$

Again by nonexpansiveness of projection,

$$|\lambda_{n+1} - \lambda|^2 \leq |\lambda_n + \eta g_n^\lambda - \lambda|^2.$$

Expand:

$$|\lambda_n - \lambda|^2 + 2\eta g_n^\lambda(\lambda_n - \lambda) + \eta^2 |g_n^\lambda|^2.$$

So

$$|\lambda_{n+1} - \lambda|^2 \leq |\lambda_n - \lambda|^2 + 2\eta g_n^\lambda(\lambda_n - \lambda) + \eta^2 G^2.$$

Rearrange:

$$2\eta g_n^\lambda(\lambda - \lambda_n) \leq |\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2 + \eta^2 G^2.$$

Divide by 2η:

$$g_n^\lambda(\lambda - \lambda_n) \leq \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 5: Convert the dual inner product into a function gap using concavity. Because the Lagrangian is concave in λ,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) \leq g_n^{\lambda}(\lambda - \lambda_n).$$

Therefore,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) \leq \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 6: Add the primal and dual inequalities. Add the inequalities from Step 3 and Step 5:

$$\begin{aligned}
&\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \\
&= \Big[\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\Big] + \Big[\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\Big] \\
&\leq \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \eta G^2.
\end{aligned}$$

Step 7: Sum from $n = 1$ to N. Summing both sides,

$$\begin{aligned}
&\sum_{n=1}^{N} \Big(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\Big) \\
&\leq \frac{\|\psi_1 - \psi\|_2^2 - \|\psi_{N+1} - \psi\|_2^2}{2\eta} + \frac{|\lambda_1 - \lambda|^2 - |\lambda_{N+1} - \lambda|^2}{2\eta} + N\eta G^2.
\end{aligned}$$

Since squared norms are nonnegative, we may drop the negative terminal terms:

$$\leq \frac{\|\psi_1 - \psi\|_2^2}{2\eta} + \frac{|\lambda_1 - \lambda|^2}{2\eta} + N\eta G^2.$$

By the diameter bounds,

$$\|\psi_1 - \psi\|_2 \leq D_\psi, \qquad |\lambda_1 - \lambda| \leq D_\lambda.$$

So

$$\sum_{n=1}^{N} \Big(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\Big) \leq \frac{D_\psi^2 + D_\lambda^2}{2\eta} + N\eta G^2 = \frac{D^2}{2\eta} + N\eta G^2.$$

Step 8: Divide by N and move to the averaged iterates. Divide by N:

$$\frac{1}{N} \sum_{n=1}^{N} \Big(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\Big) \leq \frac{D^2}{2\eta N} + \eta G^2.$$

By convexity in ψ and concavity in λ,

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) \leq \frac{1}{N} \sum_{n=1}^{N} \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda),$$

and

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \geq \frac{1}{N} \sum_{n=1}^{N} \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n).$$

Subtracting the two inequalities gives

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{D^2}{2\eta N} + \eta G^2.$$

Step 9: Choose the step size. Set

$$\eta = \frac{D}{G\sqrt{N}}.$$

Then

$$\frac{D^2}{2\eta N} = \frac{D^2}{2(D/(G\sqrt{N}))N} = \frac{DG}{2\sqrt{N}},$$

and

$$\eta G^2 = \frac{D}{G\sqrt{N}} G^2 = \frac{DG}{\sqrt{N}}.$$

Therefore

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{3DG}{2\sqrt{N}}.$$

For cleanliness, we can upper bound this by

$$\frac{2DG}{\sqrt{N}}.$$

Because the inequality holds for all $\psi \in \Psi$ and $\lambda \in [0, \Lambda]$,

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \min_{\psi \in \Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{2DG}{\sqrt{N}}.$$

This proves the saddle-gap bound.

Step 10: Convert saddle gap into a constraint-violation bound. Now define

$$g(\psi) = \widehat{\text{CVaR}}_\alpha(1 - S_z; \psi) - \varepsilon.$$

For fixed $\bar{\psi}_N$, the Lagrangian is affine in λ:

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \lambda g(\bar{\psi}_N).$$

If $g(\bar{\psi}_N) \leq 0$, then the positive part $[g(\bar{\psi}_N)]_+ = 0$, and the desired bound is trivial.

If $g(\bar{\psi}_N) > 0$, then maximizing over $\lambda \in [0, \Lambda]$ sets $\lambda = \Lambda$, so

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \Lambda g(\bar{\psi}_N).$$

In both cases,

$$\max_{\lambda\in[0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \Lambda[g(\bar{\psi}_N)]_+.$$

Subtract

$$\min_{\psi\in\Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N)$$

from both sides and use the saddle-gap bound:

$$\Lambda[g(\bar{\psi}_N)]_+ \le \frac{2DG}{\sqrt{N}}.$$

Finally divide by Λ:

$$[g(\bar{\psi}_N)]_+ \le \frac{2DG}{\Lambda\sqrt{N}}.$$

Substituting the definition of g,

$$\Big[\widehat{\text{CVaR}}_\alpha(1 - S_z; \bar{\psi}_N) - \varepsilon\Big]_+ \le \frac{2DG}{\Lambda\sqrt{N}}.$$

This proves the theorem. □

1.6 Optional appendix result: noisy-verifier DPO consistency

Proposition 1.2 (Pairwise sign consistency of verifier-noisy DPO)**.** *Consider a fixed context x_t and two actions a^+, a^-. Let the true pairwise preference label be*

$$y^\star \in \{+1, -1\},$$

where $y^\star = +1$ means a^+ is truly preferred to a^-. Suppose the verifier produces a noisy pairwise label $\widetilde{y}$ such that

$$\mathbb{P}(\widetilde{y} = y^\star \mid x_t, a^+, a^-) = 1 - \eta, \qquad \mathbb{P}(\widetilde{y} = -y^\star \mid x_t, a^+, a^-) = \eta,$$

with $\eta < 1/2$.

Let the DPO score difference be

$$u_\theta(x_t, a^+, a^-) = \beta\left[\log\frac{\pi_\theta(a^+ \mid x_t)}{\pi_{\text{ref}}(a^+ \mid x_t)} - \log\frac{\pi_\theta(a^- \mid x_t)}{\pi_{\text{ref}}(a^- \mid x_t)}\right].$$

Then the population minimizer of the noisy logistic loss

$$\mathcal{L}_{\text{DPO,noisy}}(\theta) = \mathbb{E}\Big[\log(1 + \exp(-\widetilde{y}\, u_\theta))\Big]$$

has the same sign as the true preference:

$$\text{sign}\big(u_\theta^\star(x_t, a^+, a^-)\big) = y^\star.$$

Proof. We prove the result for $y^\star = +1$. The case $y^\star = -1$ is symmetric.

Conditional on the pair (x_t, a^+, a^-), the noisy label satisfies

$$\widetilde{y} = \begin{cases} +1, & \text{with probability } 1-\eta, \\ -1, & \text{with probability } \eta. \end{cases}$$

Therefore, as a function of the scalar score u, the conditional risk is

$$R(u) = (1-\eta)\log(1+e^{-u}) + \eta\log(1+e^{u}).$$

Differentiate:

$$R'(u) = -(1-\eta)\frac{1}{1+e^u} + \eta\frac{e^u}{1+e^u}.$$

Put both terms over the common denominator $1+e^u$:

$$R'(u) = \frac{\eta e^u - (1-\eta)}{1+e^u}.$$

Set the derivative to zero:

$$\eta e^u - (1-\eta) = 0 \quad \Longleftrightarrow \quad e^u = \frac{1-\eta}{\eta}.$$

Since $\eta < 1/2$, we have

$$\frac{1-\eta}{\eta} > 1,$$

hence

$$u^\star = \log\left(\frac{1-\eta}{\eta}\right) > 0.$$

So the minimizing score is positive whenever the true preference is positive.

By symmetry, if $y^\star = -1$, then the minimizing score is negative.

Thus the optimal noisy-DPO score still has the correct sign as long as $\eta < 1/2$.

This proves the proposition. □

1.7 How to present these theorems in the paper

Remark 1.4 (Recommended placement)**.** *A clean presentation is:*

(i) Main paper: Theorem 1.1, Corollary 1.1, Theorem 1.3, and Theorem 1.4.

(ii) Main paper or appendix: Proposition 1.1.

(iii) Appendix only: Proposition 1.2.

This gives one theorem for calibration/decision theory, one for perturbation robustness, one for optimization, and one supporting proposition for verifier-guided candidate selection.